

DEVOPS MASTER COURSE

Volume 12C

Microsoft Azure Enterprise Cloud Engineering

*Landing Zones • Entra • Networking • Compute • Data • Security • Reliability •
Operations • FinOps*

Chapters 21-30 | 2026 Edition

Beginner to Enterprise Cloud Architecture and Operations

About This Part

Part C develops provider-depth in Microsoft Azure. It connects the Azure landing-zone hierarchy to Microsoft Entra identity, private networking, modern application platforms, managed data, Defender and Sentinel operations, recovery and cost engineering.

2026 platform note: The 2026 edition follows the current Azure Landing Zones model of platform and application landing zones, the Microsoft Entra naming model, managed identities, Private Link and the Azure Well-Architected and Cloud Adoption Frameworks.

Volume 12 Roadmap

Part	Coverage	Status
12A	Multi-cloud architecture and landing-zone foundations	Completed
12B	AWS enterprise cloud engineering	Course sequence
12C	Azure enterprise cloud engineering	Completed in this package
12D	Google Cloud enterprise engineering	Course sequence
12E	Hybrid and multi-cloud operations, migration, DR, security and FinOps	Course sequence
12F	Enterprise projects, certifications, interviews and final capstone	Course sequence

Scope, Safety and Cost Controls

- Use disposable sandboxes, free tiers, local simulations or approved training subscriptions. Configure budgets and expiry before creating resources.
- Never use production identities, customer data, employer account structures, internal IP ranges or confidential contracts in course artifacts.
- Cloud services, prices, limits, recommendations and certifications change. Verify official documentation before production use or exam registration.
- Run failure, security and recovery tests only in approved nonproduction environments with explicit stop conditions.
- Destroy resources, revoke credentials, remove local state copies and verify billing inventory after every lab.

Learning Outcomes

- Explain and design the major architecture domains covered by Microsoft Azure Enterprise Cloud Engineering.
- Implement provider or cross-cloud controls through version-controlled automation.
- Test identity, network, security, reliability, operations and cost behavior.
- Troubleshoot failures using effective policy, route, service and telemetry evidence.
- Produce sanitized architecture, runbook, laboratory and interview portfolio artifacts.

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Chapter 21: Azure Tenant, Management Groups and Landing Zones

Structuring platform and application landing zones at enterprise scale

Why this chapter matters: Azure governance becomes difficult when tenant, management-group, subscription and resource-group boundaries are improvised. The hierarchy must support policy inheritance, delegated ownership and durable operations.

Learning Objectives

- Explain the architecture, responsibility boundaries and major trade-offs of microsoft entra tenant and management-group hierarchy.
- Design platform versus application landing zones for enterprise scale, least privilege and bounded failure.
- Implement subscription vending and lifecycle management through version-controlled automation and policy checks.
- Troubleshoot azure policy inheritance and exceptions using provider-native telemetry and repeatable evidence.
- Deliver a sanitized portfolio artifact for design and test an azure landing-zone hierarchy.

Core Concepts

Microsoft Entra tenant and management-group hierarchy

Microsoft Entra tenant and management-group hierarchy is a design boundary, not merely a console feature. In Azure Tenant, Management Groups and Landing Zones, the team must document ownership, trust assumptions, dependency direction, failure impact and the evidence used to prove the control. The recommended implementation uses Microsoft Entra ID, Management groups, Subscriptions, Azure Landing Zones only where they satisfy an explicit requirement; equivalent services are not automatically interchangeable.

Platform versus application landing zones

A production implementation of Platform versus application landing zones separates the management plane from workload execution and constrains blast radius. Architecture reviews should evaluate quotas, regional behavior, identity propagation, private connectivity, encryption, logging and recovery. The operational contract must state who changes the control, who monitors it and how an emergency exception expires.

Subscription vending and lifecycle management

For Subscription vending and lifecycle management, repeatability matters more than a one-time successful deployment. Infrastructure as code, policy tests, deployment verification, drift detection and rollback evidence should be retained together. A design is incomplete when it works only for the original engineer or depends on undocumented console steps.

Azure Policy inheritance and exceptions

The strongest validation for Azure Policy inheritance and exceptions combines an allowed test, a denied test and a failure test. Teams should prove normal workload behavior, demonstrate that prohibited behavior is blocked and exercise recovery under a bounded fault. Metrics, logs, audit events and cost records become part of the acceptance package.

Service and Capability Map

Capability	Service or pattern	Enterprise use
Tenant	Microsoft Entra ID	Provide the identity and directory boundary for Azure.
Hierarchy	Management groups	Apply governance above subscriptions.

Capability	Service or pattern	Enterprise use
Workload boundary	Subscriptions	Separate billing, quota, policy and administration.
Landing zone	Azure Landing Zones	Standardize platform and application foundations.
Vending	Subscription vending automation	Create governed subscriptions for workload teams.
Governance	Azure Policy	Audit, deny, modify and deploy required configuration.

Architecture and Operating Decisions

- Use Azure managed services only after recording availability, security, portability and cost consequences.
- Separate production, nonproduction, security and shared services with strong administrative and billing boundaries.
- Prefer short-lived workload identity, private service access and centralized audit evidence over persistent credentials and public administration.
- Automate subscription vending and lifecycle management with reviewed plans, policy tests, staged rollout and post-deployment verification.
- Define recovery objectives, quotas, ownership and decommissioning before the service becomes business critical.

Reference Configuration or Decision Pattern



Interpretation: Treat the pattern as a starting point. Replace placeholders with measured requirements, provider limits and approved organizational standards.

Production Readiness Checklist

- ☐ Architecture diagrams and responsibility boundaries for azure tenant, management groups and landing zones are current and version controlled.
- ☐ Identity uses groups, roles, workload federation and privileged elevation with no unnecessary standing administrator access.
- ☐ Network paths, DNS behavior, encryption, audit logging and service limits are tested before production use.
- ☐ Changes are delivered through IaC with validation, policy, approval, rollback and drift detection.
- ☐ Recovery, cost allocation, ownership metadata and teardown have measurable evidence.

Troubleshooting Matrix

Symptom	Likely cause	Corrective action	Validation evidence
Deployment succeeds but service is unreachable	Route, firewall, DNS or private-endpoint path is incomplete	Trace the path hop by hop; compare route, policy, DNS and endpoint state; then run an explicit reachability test.	Successful synthetic request and flow/audit evidence
Authorization is broader than intended	Inherited role, wildcard permission or unmanaged credential	Inspect effective permissions, credential age and access logs; replace broad grants with scoped roles and short-lived identity.	Effective-access review and denied-action test

Symptom	Likely cause	Corrective action	Validation evidence
Regional failure breaks the workload	State, dependency or failover mechanism remained single-region	Map stateful dependencies, restore order and traffic controls; exercise failover and fallback against measured RTO/RPO.	Timed recovery log with data-integrity checks
Cost rises without traffic growth	Idle capacity, data transfer, logging volume or commitment mismatch	Break down spend by owner, service, region and usage driver; validate rightsizing or commitment changes against SLOs.	Cost delta plus performance and reliability validation

Hands-on Lab 21: Design and Test an Azure Landing-Zone Hierarchy

Prerequisites

- Disposable provider sandbox or a local simulation.
- Terraform or provider CLI, Git and a diagramming tool.
- Synthetic identities and data; no production credentials or confidential information.

Implementation

1. Create the architecture and responsibility model for azure tenant, management groups and landing zones using fictional enterprise requirements.
2. Implement the smallest safe sandbox using Microsoft Entra ID, Management groups, Subscriptions; use free or low-cost options where available.
3. Apply identity, network, encryption, logging, metadata and budget controls through IaC or an equivalent version-controlled simulation.
4. Run one approved behavior test and one denied behavior test; save the commands, timestamps and resulting telemetry.
5. Inject a bounded configuration or dependency failure, follow the runbook, recover and record measured recovery time.
6. Destroy or quarantine the lab, revoke temporary access and verify that no unexpected billable resources remain.

Validation Evidence

- Version-controlled architecture, IaC and decision record.
- Allowed, denied and failure-test outputs with timestamps.
- Cost, recovery, ownership and teardown verification.

Cleanup

- Destroy test resources in dependency order and verify the provider inventory and billing view.
- Revoke temporary identities, remove local state copies and sanitize all portfolio evidence.

Interview Questions and Model Answers

1. What is the most important design principle in Azure Tenant, Management Groups and Landing Zones?

Make responsibility and failure boundaries explicit, then automate controls and prove them with normal, denied and recovery tests rather than trusting a successful deployment alone.

2. How would you choose between the main Azure services in this domain?

Start with workload requirements for latency, state, scale, compliance, team capability, recovery and cost. Select the managed service whose operating model fits those requirements and document the consequences in an ADR.

3. What evidence would you require before production approval?

Architecture and threat models, reviewed IaC plan, effective-access results, network and encryption tests, known audit events, SLO and recovery validation, ownership metadata, budget routing and rollback evidence.

4. How do you reduce provider lock-in without creating a lowest-common-denominator platform?

Preserve portable source, data, interfaces and deployment artifacts where a credible exit scenario requires them, while still using provider-specific capabilities when their measurable value exceeds migration and dependency risk.

Chapter Summary

- Microsoft Entra tenant and management-group hierarchy defines a key architecture boundary.
- Platform versus application landing zones must be automated and least-privileged.
- Subscription vending and lifecycle management requires policy, telemetry and recovery evidence.
- Azure Policy inheritance and exceptions should be tested before production and after material change.

Chapter 22: Microsoft Entra Identity, RBAC and Privileged Access

Securing human and workload access with scoped, time-bound authorization

Why this chapter matters: Azure access spans directory roles, Azure RBAC, managed identities, PIM and resource-specific permissions. Effective authorization must be evaluated across every layer.

Learning Objectives

- Explain the architecture, responsibility boundaries and major trade-offs of microsoft entra federation and conditional access.
- Design azure rbac scope and custom roles for enterprise scale, least privilege and bounded failure.
- Implement privileged identity management and break-glass access through version-controlled automation and policy checks.
- Troubleshoot managed identities and workload identity federation using provider-native telemetry and repeatable evidence.
- Deliver a sanitized portfolio artifact for implement azure privileged and workload identity controls.

Core Concepts

Microsoft Entra federation and Conditional Access

Microsoft Entra federation and Conditional Access is a design boundary, not merely a console feature. In Microsoft Entra Identity, RBAC and Privileged Access, the team must document ownership, trust assumptions, dependency direction, failure impact and the evidence used to prove the control. The recommended implementation uses Microsoft Entra ID, Conditional Access, Azure RBAC, PIM only where they satisfy an explicit requirement; equivalent services are not automatically interchangeable.

Azure RBAC scope and custom roles

A production implementation of Azure RBAC scope and custom roles separates the management plane from workload execution and constrains blast radius. Architecture reviews should evaluate quotas, regional behavior, identity propagation, private connectivity, encryption, logging and recovery. The operational contract must state who changes the control, who monitors it and how an emergency exception expires.

Privileged Identity Management and break-glass access

For Privileged Identity Management and break-glass access, repeatability matters more than a one-time successful deployment. Infrastructure as code, policy tests, deployment verification, drift detection and rollback evidence should be retained together. A design is incomplete when it works only for the original engineer or depends on undocumented console steps.

Managed identities and workload identity federation

The strongest validation for Managed identities and workload identity federation combines an allowed test, a denied test and a failure test. Teams should prove normal workload behavior, demonstrate that prohibited behavior is blocked and exercise recovery under a bounded fault. Metrics, logs, audit events and cost records become part of the acceptance package.

Service and Capability Map

Capability	Service or pattern	Enterprise use
Directory access	Microsoft Entra ID	Authenticate workforce and external identities.
Conditional access	Conditional Access	Apply risk, device, location and MFA controls.

Capability	Service or pattern	Enterprise use
Resource authorization	Azure RBAC	Grant actions at management group, subscription, resource group or resource scope.
Privilege	PIM	Provide time-bound eligible administrative access.
Workload identity	Managed identities / workload identity federation	Issue tokens without stored application secrets.
Secrets	Azure Key Vault	Protect keys, secrets and certificates.

Architecture and Operating Decisions

- Use Azure managed services only after recording availability, security, portability and cost consequences.
- Separate production, nonproduction, security and shared services with strong administrative and billing boundaries.
- Prefer short-lived workload identity, private service access and centralized audit evidence over persistent credentials and public administration.
- Automate privileged identity management and break-glass access with reviewed plans, policy tests, staged rollout and post-deployment verification.
- Define recovery objectives, quotas, ownership and decommissioning before the service becomes business critical.

Reference Configuration or Decision Pattern

```
identity_standard:
  humans: entra_groups_plus_pim
  workloads: managed_identity_or_federated_credential
  emergency: two_cloud_only_break_glass_accounts
  directory_roles: eligible_not_permanent
  azure_roles: least_scope
  access_reviews: quarterly
  validation: [conditional_access, denied_action, token_claims, audit_logs]
```

Interpretation: Treat the pattern as a starting point. Replace placeholders with measured requirements, provider limits and approved organizational standards.

Production Readiness Checklist

- ☐ Architecture diagrams and responsibility boundaries for microsoft entra identity, rbac and privileged access are current and version controlled.
- ☐ Identity uses groups, roles, workload federation and privileged elevation with no unnecessary standing administrator access.
- ☐ Network paths, DNS behavior, encryption, audit logging and service limits are tested before production use.
- ☐ Changes are delivered through IaC with validation, policy, approval, rollback and drift detection.
- ☐ Recovery, cost allocation, ownership metadata and teardown have measurable evidence.

Troubleshooting Matrix

Symptom	Likely cause	Corrective action	Validation evidence
Deployment succeeds but service is unreachable	Route, firewall, DNS or private-endpoint path is incomplete	Trace the path hop by hop; compare route, policy, DNS and endpoint state; then run an explicit reachability test.	Successful synthetic request and flow/audit evidence
Authorization is broader than intended	Inherited role, wildcard permission or unmanaged credential	Inspect effective permissions, credential age and access logs; replace broad grants with scoped roles and short-lived identity.	Effective-access review and denied-action test
Regional failure breaks the workload	State, dependency or failover mechanism remained single-region	Map stateful dependencies, restore order and traffic controls; exercise failover and fallback against measured RTO/RPO.	Timed recovery log with data-integrity checks

Symptom	Likely cause	Corrective action	Validation evidence
Cost rises without traffic growth	Idle capacity, data transfer, logging volume or commitment mismatch	Break down spend by owner, service, region and usage driver; validate rightsizing or commitment changes against SLOs.	Cost delta plus performance and reliability validation

Hands-on Lab 22: Implement Azure Privileged and Workload Identity Controls

Prerequisites

- Disposable provider sandbox or a local simulation.
- Terraform or provider CLI, Git and a diagramming tool.
- Synthetic identities and data; no production credentials or confidential information.

Implementation

1. Create the architecture and responsibility model for microsoft entra identity, rbac and privileged access using fictional enterprise requirements.
2. Implement the smallest safe sandbox using Microsoft Entra ID, Conditional Access, Azure RBAC; use free or low-cost options where available.
3. Apply identity, network, encryption, logging, metadata and budget controls through IaC or an equivalent version-controlled simulation.
4. Run one approved behavior test and one denied behavior test; save the commands, timestamps and resulting telemetry.
5. Inject a bounded configuration or dependency failure, follow the runbook, recover and record measured recovery time.
6. Destroy or quarantine the lab, revoke temporary access and verify that no unexpected billable resources remain.

Validation Evidence

- Version-controlled architecture, IaC and decision record.
- Allowed, denied and failure-test outputs with timestamps.
- Cost, recovery, ownership and teardown verification.

Cleanup

- Destroy test resources in dependency order and verify the provider inventory and billing view.
- Revoke temporary identities, remove local state copies and sanitize all portfolio evidence.

Interview Questions and Model Answers

1. What is the most important design principle in Microsoft Entra Identity, RBAC and Privileged Access?

Make responsibility and failure boundaries explicit, then automate controls and prove them with normal, denied and recovery tests rather than trusting a successful deployment alone.

2. How would you choose between the main Azure services in this domain?

Start with workload requirements for latency, state, scale, compliance, team capability, recovery and cost. Select the managed service whose operating model fits those requirements and document the consequences in an ADR.

3. What evidence would you require before production approval?

Architecture and threat models, reviewed IaC plan, effective-access results, network and encryption tests, known audit events, SLO and recovery validation, ownership metadata, budget routing and rollback evidence.

4. How do you reduce provider lock-in without creating a lowest-common-denominator platform?

Preserve portable source, data, interfaces and deployment artifacts where a credible exit scenario requires them, while still using provider-specific capabilities when their measurable value exceeds migration and dependency risk.

Chapter Summary

- Microsoft Entra federation and Conditional Access defines a key architecture boundary.
- Azure RBAC scope and custom roles must be automated and least-privileged.
- Privileged Identity Management and break-glass access requires policy, telemetry and recovery evidence.
- Managed identities and workload identity federation should be tested before production and after material change.

Chapter 23: Azure Networking, Private Connectivity and Hybrid Design

Engineering hub-spoke or Virtual WAN connectivity with predictable DNS

Why this chapter matters: Azure networks require coordinated address planning, routing, private endpoints, DNS and security. Peering alone does not create transitive routing or a governed enterprise topology.

Learning Objectives

- Explain the architecture, responsibility boundaries and major trade-offs of virtual networks, ipam and segmentation.
- Design hub-spoke and azure virtual wan for enterprise scale, least privilege and bounded failure.
- Implement expressroute, vpn and hybrid routing through version-controlled automation and policy checks.
- Troubleshoot private link, dns and controlled ingress/egress using provider-native telemetry and repeatable evidence.
- Deliver a sanitized portfolio artifact for build an azure hub-spoke network with private endpoints.

Core Concepts

Virtual networks, IPAM and segmentation

Virtual networks, IPAM and segmentation is a design boundary, not merely a console feature. In Azure Networking, Private Connectivity and Hybrid Design, the team must document ownership, trust assumptions, dependency direction, failure impact and the evidence used to prove the control. The recommended implementation uses Azure Virtual Network, Hub-spoke / Virtual WAN, ExpressRoute / VPN Gateway, Private Link / private endpoints only where they satisfy an explicit requirement; equivalent services are not automatically interchangeable.

Hub-spoke and Azure Virtual WAN

A production implementation of Hub-spoke and Azure Virtual WAN separates the management plane from workload execution and constrains blast radius. Architecture reviews should evaluate quotas, regional behavior, identity propagation, private connectivity, encryption, logging and recovery. The operational contract must state who changes the control, who monitors it and how an emergency exception expires.

ExpressRoute, VPN and hybrid routing

For ExpressRoute, VPN and hybrid routing, repeatability matters more than a one-time successful deployment. Infrastructure as code, policy tests, deployment verification, drift detection and rollback evidence should be retained together. A design is incomplete when it works only for the original engineer or depends on undocumented console steps.

Private Link, DNS and controlled ingress/egress

The strongest validation for Private Link, DNS and controlled ingress/egress combines an allowed test, a denied test and a failure test. Teams should prove normal workload behavior, demonstrate that prohibited behavior is blocked and exercise recovery under a bounded fault. Metrics, logs, audit events and cost records become part of the acceptance package.

Service and Capability Map

Capability	Service or pattern	Enterprise use
Network boundary	Azure Virtual Network	Segment workloads into subnets and security domains.
Transit	Hub-spoke / Virtual WAN	Centralize connectivity and shared network services.

Capability	Service or pattern	Enterprise use
Hybrid	ExpressRoute / VPN Gateway	Connect on-premises with private or encrypted paths.
Private PaaS	Private Link / private endpoints	Access Azure services through private IPs.
DNS	Azure DNS / Private DNS Resolver	Operate public, private and hybrid name resolution.
Security	Azure Firewall / NSG	Control network flows and centralized inspection.

Architecture and Operating Decisions

- Use Azure managed services only after recording availability, security, portability and cost consequences.
- Separate production, nonproduction, security and shared services with strong administrative and billing boundaries.
- Prefer short-lived workload identity, private service access and centralized audit evidence over persistent credentials and public administration.
- Automate expressroute, vpn and hybrid routing with reviewed plans, policy tests, staged rollout and post-deployment verification.
- Define recovery objectives, quotas, ownership and decommissioning before the service becomes business critical.

Reference Configuration or Decision Pattern

```
network:
  topology: virtual_wan_or_hub_spoke_by_scale
  address_plan: central_ipam
  hybrid: dual_expressroute_plus_vpn_backup
  paas_access: private_endpoints_default
  dns: private_dns_resolver_linked_zones
  egress: centralized_with_fqdn_policy
  validation: [network_watcher, connection_monitor, flow_logs, dns_tests]
```

Interpretation: Treat the pattern as a starting point. Replace placeholders with measured requirements, provider limits and approved organizational standards.

Production Readiness Checklist

- ☐ Architecture diagrams and responsibility boundaries for azure networking, private connectivity and hybrid design are current and version controlled.
- ☐ Identity uses groups, roles, workload federation and privileged elevation with no unnecessary standing administrator access.
- ☐ Network paths, DNS behavior, encryption, audit logging and service limits are tested before production use.
- ☐ Changes are delivered through IaC with validation, policy, approval, rollback and drift detection.
- ☐ Recovery, cost allocation, ownership metadata and teardown have measurable evidence.

Troubleshooting Matrix

Symptom	Likely cause	Corrective action	Validation evidence
Deployment succeeds but service is unreachable	Route, firewall, DNS or private-endpoint path is incomplete	Trace the path hop by hop; compare route, policy, DNS and endpoint state; then run an explicit reachability test.	Successful synthetic request and flow/audit evidence
Authorization is broader than intended	Inherited role, wildcard permission or unmanaged credential	Inspect effective permissions, credential age and access logs; replace broad grants with scoped roles and short-lived identity.	Effective-access review and denied-action test
Regional failure breaks the workload	State, dependency or failover mechanism remained single-region	Map stateful dependencies, restore order and traffic controls; exercise failover and fallback against measured RTO/RPO.	Timed recovery log with data-integrity checks

Symptom	Likely cause	Corrective action	Validation evidence
Cost rises without traffic growth	Idle capacity, data transfer, logging volume or commitment mismatch	Break down spend by owner, service, region and usage driver; validate rightsizing or commitment changes against SLOs.	Cost delta plus performance and reliability validation

Hands-on Lab 23: Build an Azure Hub-Spoke Network with Private Endpoints

Prerequisites

- Disposable provider sandbox or a local simulation.
- Terraform or provider CLI, Git and a diagramming tool.
- Synthetic identities and data; no production credentials or confidential information.

Implementation

1. Create the architecture and responsibility model for azure networking, private connectivity and hybrid design using fictional enterprise requirements.
2. Implement the smallest safe sandbox using Azure Virtual Network, Hub-spoke / Virtual WAN, ExpressRoute / VPN Gateway; use free or low-cost options where available.
3. Apply identity, network, encryption, logging, metadata and budget controls through IaC or an equivalent version-controlled simulation.
4. Run one approved behavior test and one denied behavior test; save the commands, timestamps and resulting telemetry.
5. Inject a bounded configuration or dependency failure, follow the runbook, recover and record measured recovery time.
6. Destroy or quarantine the lab, revoke temporary access and verify that no unexpected billable resources remain.

Validation Evidence

- Version-controlled architecture, IaC and decision record.
- Allowed, denied and failure-test outputs with timestamps.
- Cost, recovery, ownership and teardown verification.

Cleanup

- Destroy test resources in dependency order and verify the provider inventory and billing view.
- Revoke temporary identities, remove local state copies and sanitize all portfolio evidence.

Interview Questions and Model Answers

1. What is the most important design principle in Azure Networking, Private Connectivity and Hybrid Design?

Make responsibility and failure boundaries explicit, then automate controls and prove them with normal, denied and recovery tests rather than trusting a successful deployment alone.

2. How would you choose between the main Azure services in this domain?

Start with workload requirements for latency, state, scale, compliance, team capability, recovery and cost. Select the managed service whose operating model fits those requirements and document the consequences in an ADR.

3. What evidence would you require before production approval?

Architecture and threat models, reviewed IaC plan, effective-access results, network and encryption tests, known audit events, SLO and recovery validation, ownership metadata, budget routing and rollback evidence.

4. How do you reduce provider lock-in without creating a lowest-common-denominator platform?

Preserve portable source, data, interfaces and deployment artifacts where a credible exit scenario requires them, while still using provider-specific capabilities when their measurable value exceeds migration and dependency risk.

Chapter Summary

- Virtual networks, IPAM and segmentation defines a key architecture boundary.
- Hub-spoke and Azure Virtual WAN must be automated and least-privileged.
- ExpressRoute, VPN and hybrid routing requires policy, telemetry and recovery evidence.
- Private Link, DNS and controlled ingress/egress should be tested before production and after material change.

Chapter 24: Azure Compute, Containers and Serverless Architecture

Choosing VMs, App Service, Functions, Container Apps and AKS

Why this chapter matters: Azure compute platforms trade control for operational abstraction. Placement should reflect runtime requirements, scaling, networking, deployment, compliance and team capability.

Learning Objectives

- Explain the architecture, responsibility boundaries and major trade-offs of virtual machines and virtual machine scale sets.
- Design app service, functions and container apps for enterprise scale, least privilege and bounded failure.
- Implement azure kubernetes service architecture through version-controlled automation and policy checks.
- Troubleshoot images, autoscaling and deployment safety using provider-native telemetry and repeatable evidence.
- Deliver a sanitized portfolio artifact for compare azure compute placement with a deployable workload.

Core Concepts

Virtual Machines and Virtual Machine Scale Sets

Virtual Machines and Virtual Machine Scale Sets is a design boundary, not merely a console feature. In Azure Compute, Containers and Serverless Architecture, the team must document ownership, trust assumptions, dependency direction, failure impact and the evidence used to prove the control. The recommended implementation uses Azure VMs / VM Scale Sets, Azure App Service, Azure Functions, Azure Container Apps only where they satisfy an explicit requirement; equivalent services are not automatically interchangeable.

App Service, Functions and Container Apps

A production implementation of App Service, Functions and Container Apps separates the management plane from workload execution and constrains blast radius. Architecture reviews should evaluate quotas, regional behavior, identity propagation, private connectivity, encryption, logging and recovery. The operational contract must state who changes the control, who monitors it and how an emergency exception expires.

Azure Kubernetes Service architecture

For Azure Kubernetes Service architecture, repeatability matters more than a one-time successful deployment. Infrastructure as code, policy tests, deployment verification, drift detection and rollback evidence should be retained together. A design is incomplete when it works only for the original engineer or depends on undocumented console steps.

Images, autoscaling and deployment safety

The strongest validation for Images, autoscaling and deployment safety combines an allowed test, a denied test and a failure test. Teams should prove normal workload behavior, demonstrate that prohibited behavior is blocked and exercise recovery under a bounded fault. Metrics, logs, audit events and cost records become part of the acceptance package.

Service and Capability Map

Capability	Service or pattern	Enterprise use
Virtual machines	Azure VMs / VM Scale Sets	Run customizable and legacy compute.
Web platform	Azure App Service	Host managed web applications and APIs.
Functions	Azure Functions	Run event-driven functions with managed scaling.

Capability	Service or pattern	Enterprise use
Serverless containers	Azure Container Apps	Run microservices and jobs without cluster administration.
Kubernetes	AKS	Operate Kubernetes workloads with managed control plane.
Images	Azure Compute Gallery / ACR	Distribute governed machine and container images.

Architecture and Operating Decisions

- Use Azure managed services only after recording availability, security, portability and cost consequences.
- Separate production, nonproduction, security and shared services with strong administrative and billing boundaries.
- Prefer short-lived workload identity, private service access and centralized audit evidence over persistent credentials and public administration.
- Automate azure kubernetes service architecture with reviewed plans, policy tests, staged rollout and post-deployment verification.
- Define recovery objectives, quotas, ownership and decommissioning before the service becomes business critical.

Reference Configuration or Decision Pattern

```
placement:
  legacy_or_custom_os: vm_scale_sets
  managed_web: app_service
  event_processing: functions
  serverless_microservices: container_apps
  kubernetes_requirements: aks
controls: [availability_zones, managed_identity, autoscale, image_scan, rollback]
```

Interpretation: Treat the pattern as a starting point. Replace placeholders with measured requirements, provider limits and approved organizational standards.

Production Readiness Checklist

- ☐ Architecture diagrams and responsibility boundaries for azure compute, containers and serverless architecture are current and version controlled.
- ☐ Identity uses groups, roles, workload federation and privileged elevation with no unnecessary standing administrator access.
- ☐ Network paths, DNS behavior, encryption, audit logging and service limits are tested before production use.
- ☐ Changes are delivered through IaC with validation, policy, approval, rollback and drift detection.
- ☐ Recovery, cost allocation, ownership metadata and teardown have measurable evidence.

Troubleshooting Matrix

Symptom	Likely cause	Corrective action	Validation evidence
Deployment succeeds but service is unreachable	Route, firewall, DNS or private-endpoint path is incomplete	Trace the path hop by hop; compare route, policy, DNS and endpoint state; then run an explicit reachability test.	Successful synthetic request and flow/audit evidence
Authorization is broader than intended	Inherited role, wildcard permission or unmanaged credential	Inspect effective permissions, credential age and access logs; replace broad grants with scoped roles and short-lived identity.	Effective-access review and denied-action test
Regional failure breaks the workload	State, dependency or failover mechanism remained single-region	Map stateful dependencies, restore order and traffic controls; exercise failover and fallback against measured RTO/RPO.	Timed recovery log with data-integrity checks

Symptom	Likely cause	Corrective action	Validation evidence
Cost rises without traffic growth	Idle capacity, data transfer, logging volume or commitment mismatch	Break down spend by owner, service, region and usage driver; validate rightsizing or commitment changes against SLOs.	Cost delta plus performance and reliability validation

Hands-on Lab 24: Compare Azure Compute Placement with a Deployable Workload

Prerequisites

- Disposable provider sandbox or a local simulation.
- Terraform or provider CLI, Git and a diagramming tool.
- Synthetic identities and data; no production credentials or confidential information.

Implementation

1. Create the architecture and responsibility model for azure compute, containers and serverless architecture using fictional enterprise requirements.
2. Implement the smallest safe sandbox using Azure VMs / VM Scale Sets, Azure App Service, Azure Functions; use free or low-cost options where available.
3. Apply identity, network, encryption, logging, metadata and budget controls through IaC or an equivalent version-controlled simulation.
4. Run one approved behavior test and one denied behavior test; save the commands, timestamps and resulting telemetry.
5. Inject a bounded configuration or dependency failure, follow the runbook, recover and record measured recovery time.
6. Destroy or quarantine the lab, revoke temporary access and verify that no unexpected billable resources remain.

Validation Evidence

- Version-controlled architecture, IaC and decision record.
- Allowed, denied and failure-test outputs with timestamps.
- Cost, recovery, ownership and teardown verification.

Cleanup

- Destroy test resources in dependency order and verify the provider inventory and billing view.
- Revoke temporary identities, remove local state copies and sanitize all portfolio evidence.

Interview Questions and Model Answers

1. What is the most important design principle in Azure Compute, Containers and Serverless Architecture?

Make responsibility and failure boundaries explicit, then automate controls and prove them with normal, denied and recovery tests rather than trusting a successful deployment alone.

2. How would you choose between the main Azure services in this domain?

Start with workload requirements for latency, state, scale, compliance, team capability, recovery and cost. Select the managed service whose operating model fits those requirements and document the consequences in an ADR.

3. What evidence would you require before production approval?

Architecture and threat models, reviewed IaC plan, effective-access results, network and encryption tests, known audit events, SLO and recovery validation, ownership metadata, budget routing and rollback evidence.

4. How do you reduce provider lock-in without creating a lowest-common-denominator platform?

Preserve portable source, data, interfaces and deployment artifacts where a credible exit scenario requires them, while still using provider-specific capabilities when their measurable value exceeds migration and dependency risk.

Chapter Summary

- Virtual Machines and Virtual Machine Scale Sets defines a key architecture boundary.
- App Service, Functions and Container Apps must be automated and least-privileged.
- Azure Kubernetes Service architecture requires policy, telemetry and recovery evidence.
- Images, autoscaling and deployment safety should be tested before production and after material change.

Chapter 25: Azure Storage, Databases and Data Protection

Designing storage and data services for access, consistency and recovery

Why this chapter matters: Azure data architecture must explicitly define redundancy, encryption, network access, retention, restore order and regional behavior. Defaults are not a recovery plan.

Learning Objectives

- Explain the architecture, responsibility boundaries and major trade-offs of storage account redundancy and network controls.
- Design managed disks and azure files for enterprise scale, least privilege and bounded failure.
- Implement azure sql, postgresql and cosmos db through version-controlled automation and policy checks.
- Troubleshoot backup, site recovery and restore validation using provider-native telemetry and repeatable evidence.
- Deliver a sanitized portfolio artifact for implement azure data protection and restore evidence.

Core Concepts

Storage account redundancy and network controls

Storage account redundancy and network controls is a design boundary, not merely a console feature. In Azure Storage, Databases and Data Protection, the team must document ownership, trust assumptions, dependency direction, failure impact and the evidence used to prove the control. The recommended implementation uses Blob Storage / ADLS Gen2, Managed Disks, Azure Files / Azure NetApp Files, Azure SQL / Azure Database for PostgreSQL only where they satisfy an explicit requirement; equivalent services are not automatically interchangeable.

Managed disks and Azure Files

A production implementation of Managed disks and Azure Files separates the management plane from workload execution and constrains blast radius. Architecture reviews should evaluate quotas, regional behavior, identity propagation, private connectivity, encryption, logging and recovery. The operational contract must state who changes the control, who monitors it and how an emergency exception expires.

Azure SQL, PostgreSQL and Cosmos DB

For Azure SQL, PostgreSQL and Cosmos DB, repeatability matters more than a one-time successful deployment. Infrastructure as code, policy tests, deployment verification, drift detection and rollback evidence should be retained together. A design is incomplete when it works only for the original engineer or depends on undocumented console steps.

Backup, Site Recovery and restore validation

The strongest validation for Backup, Site Recovery and restore validation combines an allowed test, a denied test and a failure test. Teams should prove normal workload behavior, demonstrate that prohibited behavior is blocked and exercise recovery under a bounded fault. Metrics, logs, audit events and cost records become part of the acceptance package.

Service and Capability Map

Capability	Service or pattern	Enterprise use
Object	Blob Storage / ADLS Gen2	Store objects, archives and analytical data.
Block	Managed Disks	Provide persistent disks for Azure VMs.
File	Azure Files / Azure NetApp Files	Offer managed SMB/NFS file services.
Relational	Azure SQL / Azure Database for PostgreSQL	Run managed relational workloads.

Capability	Service or pattern	Enterprise use
Globally distributed	Azure Cosmos DB	Provide multi-model data with configurable consistency.
Protection	Azure Backup	Centralize backup policy and restore operations.

Architecture and Operating Decisions

- Use Azure managed services only after recording availability, security, portability and cost consequences.
- Separate production, nonproduction, security and shared services with strong administrative and billing boundaries.
- Prefer short-lived workload identity, private service access and centralized audit evidence over persistent credentials and public administration.
- Automate azure sql, postgresql and cosmos db with reviewed plans, policy tests, staged rollout and post-deployment verification.
- Define recovery objectives, quotas, ownership and decommissioning before the service becomes business critical.

Reference Configuration or Decision Pattern

```
data_policy:
  classification: confidential
  storage_redundancy: choose_by_rpo_and_region_requirement
  public_network_access: disabled_by_default
  encryption: platform_or_customer_managed_key
  backups: policy_plus_immutable_vault_where_required
  restore_test: quarterly
  data_exit: documented_export_format
```

Interpretation: Treat the pattern as a starting point. Replace placeholders with measured requirements, provider limits and approved organizational standards.

Production Readiness Checklist

- ☐ Architecture diagrams and responsibility boundaries for azure storage, databases and data protection are current and version controlled.
- ☐ Identity uses groups, roles, workload federation and privileged elevation with no unnecessary standing administrator access.
- ☐ Network paths, DNS behavior, encryption, audit logging and service limits are tested before production use.
- ☐ Changes are delivered through IaC with validation, policy, approval, rollback and drift detection.
- ☐ Recovery, cost allocation, ownership metadata and teardown have measurable evidence.

Troubleshooting Matrix

Symptom	Likely cause	Corrective action	Validation evidence
Deployment succeeds but service is unreachable	Route, firewall, DNS or private-endpoint path is incomplete	Trace the path hop by hop; compare route, policy, DNS and endpoint state; then run an explicit reachability test.	Successful synthetic request and flow/audit evidence
Authorization is broader than intended	Inherited role, wildcard permission or unmanaged credential	Inspect effective permissions, credential age and access logs; replace broad grants with scoped roles and short-lived identity.	Effective-access review and denied-action test
Regional failure breaks the workload	State, dependency or failover mechanism remained single-region	Map stateful dependencies, restore order and traffic controls; exercise failover and fallback against measured RTO/RPO.	Timed recovery log with data-integrity checks
Cost rises without traffic growth	Idle capacity, data transfer, logging volume or commitment mismatch	Break down spend by owner, service, region and usage driver; validate rightsizing or commitment changes against SLOs.	Cost delta plus performance and reliability validation

Hands-on Lab 25: Implement Azure Data Protection and Restore Evidence

Prerequisites

- Disposable provider sandbox or a local simulation.
- Terraform or provider CLI, Git and a diagramming tool.
- Synthetic identities and data; no production credentials or confidential information.

Implementation

1. Create the architecture and responsibility model for azure storage, databases and data protection using fictional enterprise requirements.
2. Implement the smallest safe sandbox using Blob Storage / ADLS Gen2, Managed Disks, Azure Files / Azure NetApp Files; use free or low-cost options where available.
3. Apply identity, network, encryption, logging, metadata and budget controls through IaC or an equivalent version-controlled simulation.
4. Run one approved behavior test and one denied behavior test; save the commands, timestamps and resulting telemetry.
5. Inject a bounded configuration or dependency failure, follow the runbook, recover and record measured recovery time.
6. Destroy or quarantine the lab, revoke temporary access and verify that no unexpected billable resources remain.

Validation Evidence

- Version-controlled architecture, IaC and decision record.
- Allowed, denied and failure-test outputs with timestamps.
- Cost, recovery, ownership and teardown verification.

Cleanup

- Destroy test resources in dependency order and verify the provider inventory and billing view.
- Revoke temporary identities, remove local state copies and sanitize all portfolio evidence.

Interview Questions and Model Answers

1. What is the most important design principle in Azure Storage, Databases and Data Protection?

Make responsibility and failure boundaries explicit, then automate controls and prove them with normal, denied and recovery tests rather than trusting a successful deployment alone.

2. How would you choose between the main Azure services in this domain?

Start with workload requirements for latency, state, scale, compliance, team capability, recovery and cost. Select the managed service whose operating model fits those requirements and document the consequences in an ADR.

3. What evidence would you require before production approval?

Architecture and threat models, reviewed IaC plan, effective-access results, network and encryption tests, known audit events, SLO and recovery validation, ownership metadata, budget routing and rollback evidence.

4. How do you reduce provider lock-in without creating a lowest-common-denominator platform?

Preserve portable source, data, interfaces and deployment artifacts where a credible exit scenario requires them, while still using provider-specific capabilities when their measurable value exceeds migration and dependency risk.

Chapter Summary

- Storage account redundancy and network controls defines a key architecture boundary.
- Managed disks and Azure Files must be automated and least-privileged.
- Azure SQL, PostgreSQL and Cosmos DB requires policy, telemetry and recovery evidence.
- Backup, Site Recovery and restore validation should be tested before production and after material change.

Chapter 26: Azure Security, Defender and Sentinel Operations

Integrating posture management, protection and security analytics

Why this chapter matters: Security posture findings must become owned remediation and detections must connect to investigation and containment. Central tools do not replace workload security engineering.

Learning Objectives

- Explain the architecture, responsibility boundaries and major trade-offs of defender for cloud posture and workload protection.
- Design microsoft sentinel analytics and response for enterprise scale, least privilege and bounded failure.
- Implement key vault, encryption and secret governance through version-controlled automation and policy checks.
- Troubleshoot azure firewall, waf, ddos and policy controls using provider-native telemetry and repeatable evidence.
- Deliver a sanitized portfolio artifact for build an azure security posture and incident workflow.

Core Concepts

Defender for Cloud posture and workload protection

Defender for Cloud posture and workload protection is a design boundary, not merely a console feature. In Azure Security, Defender and Sentinel Operations, the team must document ownership, trust assumptions, dependency direction, failure impact and the evidence used to prove the control. The recommended implementation uses Microsoft Defender for Cloud, Microsoft Sentinel, Azure Key Vault / Managed HSM, Azure Policy only where they satisfy an explicit requirement; equivalent services are not automatically interchangeable.

Microsoft Sentinel analytics and response

A production implementation of Microsoft Sentinel analytics and response separates the management plane from workload execution and constrains blast radius. Architecture reviews should evaluate quotas, regional behavior, identity propagation, private connectivity, encryption, logging and recovery. The operational contract must state who changes the control, who monitors it and how an emergency exception expires.

Key Vault, encryption and secret governance

For Key Vault, encryption and secret governance, repeatability matters more than a one-time successful deployment. Infrastructure as code, policy tests, deployment verification, drift detection and rollback evidence should be retained together. A design is incomplete when it works only for the original engineer or depends on undocumented console steps.

Azure Firewall, WAF, DDoS and policy controls

The strongest validation for Azure Firewall, WAF, DDoS and policy controls combines an allowed test, a denied test and a failure test. Teams should prove normal workload behavior, demonstrate that prohibited behavior is blocked and exercise recovery under a bounded fault. Metrics, logs, audit events and cost records become part of the acceptance package.

Service and Capability Map

Capability	Service or pattern	Enterprise use
Posture	Microsoft Defender for Cloud	Assess recommendations and workload protection.
SIEM/SOAR	Microsoft Sentinel	Correlate security data and automate response.
Secrets and keys	Azure Key Vault / Managed HSM	Protect cryptographic material and application secrets.

Capability	Service or pattern	Enterprise use
Governance	Azure Policy	Enforce or audit resource standards.
Network security	Azure Firewall / WAF / DDoS Protection	Protect network and application edges.
Exposure	Defender External Attack Surface Management	Discover externally visible assets.

Architecture and Operating Decisions

- Use Azure managed services only after recording availability, security, portability and cost consequences.
- Separate production, nonproduction, security and shared services with strong administrative and billing boundaries.
- Prefer short-lived workload identity, private service access and centralized audit evidence over persistent credentials and public administration.
- Automate key vault, encryption and secret governance with reviewed plans, policy tests, staged rollout and post-deployment verification.
- Define recovery objectives, quotas, ownership and decommissioning before the service becomes business critical.

Reference Configuration or Decision Pattern

```
security_operations:
  policy: initiative_assignments_at_management_group
  defender: enabled_by_workload_risk
  sentinel: central_workspace_with_data_governance
  key_vault: purge_protection_and_private_endpoint
  response:
    high_severity: isolate_identity_or_resource
  evidence: activity_logs_and_sentinel_timeline
```

Interpretation: Treat the pattern as a starting point. Replace placeholders with measured requirements, provider limits and approved organizational standards.

Production Readiness Checklist

- ☐ Architecture diagrams and responsibility boundaries for azure security, defender and sentinel operations are current and version controlled.
- ☐ Identity uses groups, roles, workload federation and privileged elevation with no unnecessary standing administrator access.
- ☐ Network paths, DNS behavior, encryption, audit logging and service limits are tested before production use.
- ☐ Changes are delivered through IaC with validation, policy, approval, rollback and drift detection.
- ☐ Recovery, cost allocation, ownership metadata and teardown have measurable evidence.

Troubleshooting Matrix

Symptom	Likely cause	Corrective action	Validation evidence
Deployment succeeds but service is unreachable	Route, firewall, DNS or private-endpoint path is incomplete	Trace the path hop by hop; compare route, policy, DNS and endpoint state; then run an explicit reachability test.	Successful synthetic request and flow/audit evidence
Authorization is broader than intended	Inherited role, wildcard permission or unmanaged credential	Inspect effective permissions, credential age and access logs; replace broad grants with scoped roles and short-lived identity.	Effective-access review and denied-action test
Regional failure breaks the workload	State, dependency or failover mechanism remained single-region	Map stateful dependencies, restore order and traffic controls; exercise failover and fallback against measured RTO/RPO.	Timed recovery log with data-integrity checks
Cost rises without traffic growth	Idle capacity, data transfer, logging volume or commitment mismatch	Break down spend by owner, service, region and usage driver; validate rightsizing or commitment changes against SLOs.	Cost delta plus performance and reliability validation

Hands-on Lab 26: Build an Azure Security Posture and Incident Workflow

Prerequisites

- Disposable provider sandbox or a local simulation.
- Terraform or provider CLI, Git and a diagramming tool.
- Synthetic identities and data; no production credentials or confidential information.

Implementation

1. Create the architecture and responsibility model for azure security, defender and sentinel operations using fictional enterprise requirements.
2. Implement the smallest safe sandbox using Microsoft Defender for Cloud, Microsoft Sentinel, Azure Key Vault / Managed HSM; use free or low-cost options where available.
3. Apply identity, network, encryption, logging, metadata and budget controls through IaC or an equivalent version-controlled simulation.
4. Run one approved behavior test and one denied behavior test; save the commands, timestamps and resulting telemetry.
5. Inject a bounded configuration or dependency failure, follow the runbook, recover and record measured recovery time.
6. Destroy or quarantine the lab, revoke temporary access and verify that no unexpected billable resources remain.

Validation Evidence

- Version-controlled architecture, IaC and decision record.
- Allowed, denied and failure-test outputs with timestamps.
- Cost, recovery, ownership and teardown verification.

Cleanup

- Destroy test resources in dependency order and verify the provider inventory and billing view.
- Revoke temporary identities, remove local state copies and sanitize all portfolio evidence.

Interview Questions and Model Answers

1. What is the most important design principle in Azure Security, Defender and Sentinel Operations?

Make responsibility and failure boundaries explicit, then automate controls and prove them with normal, denied and recovery tests rather than trusting a successful deployment alone.

2. How would you choose between the main Azure services in this domain?

Start with workload requirements for latency, state, scale, compliance, team capability, recovery and cost. Select the managed service whose operating model fits those requirements and document the consequences in an ADR.

3. What evidence would you require before production approval?

Architecture and threat models, reviewed IaC plan, effective-access results, network and encryption tests, known audit events, SLO and recovery validation, ownership metadata, budget routing and rollback evidence.

4. How do you reduce provider lock-in without creating a lowest-common-denominator platform?

Preserve portable source, data, interfaces and deployment artifacts where a credible exit scenario requires them, while still using provider-specific capabilities when their measurable value exceeds migration and dependency risk.

Chapter Summary

- Defender for Cloud posture and workload protection defines a key architecture boundary.
- Microsoft Sentinel analytics and response must be automated and least-privileged.
- Key Vault, encryption and secret governance requires policy, telemetry and recovery evidence.
- Azure Firewall, WAF, DDoS and policy controls should be tested before production and after material change.

Chapter 27: Azure Reliability, Global Traffic and Disaster Recovery

Using zones, paired regions and tested recovery patterns

Why this chapter matters: Azure availability features have service-specific behavior. The architecture must model dependency failure, data replication, traffic control and failback, not only select a secondary region.

Learning Objectives

- Explain the architecture, responsibility boundaries and major trade-offs of availability zones and zone-redundant services.
- Design azure front door, traffic manager and global routing for enterprise scale, least privilege and bounded failure.
- Implement azure site recovery and backup-based recovery through version-controlled automation and policy checks.
- Troubleshoot multi-region data and dependency design using provider-native telemetry and repeatable evidence.
- Deliver a sanitized portfolio artifact for run an azure regional failover and failback exercise.

Core Concepts

Availability zones and zone-redundant services

Availability zones and zone-redundant services is a design boundary, not merely a console feature. In Azure Reliability, Global Traffic and Disaster Recovery, the team must document ownership, trust assumptions, dependency direction, failure impact and the evidence used to prove the control. The recommended implementation uses Availability Zones, Azure Front Door, Traffic Manager, Azure Site Recovery only where they satisfy an explicit requirement; equivalent services are not automatically interchangeable.

Azure Front Door, Traffic Manager and global routing

A production implementation of Azure Front Door, Traffic Manager and global routing separates the management plane from workload execution and constrains blast radius. Architecture reviews should evaluate quotas, regional behavior, identity propagation, private connectivity, encryption, logging and recovery. The operational contract must state who changes the control, who monitors it and how an emergency exception expires.

Azure Site Recovery and backup-based recovery

For Azure Site Recovery and backup-based recovery, repeatability matters more than a one-time successful deployment. Infrastructure as code, policy tests, deployment verification, drift detection and rollback evidence should be retained together. A design is incomplete when it works only for the original engineer or depends on undocumented console steps.

Multi-region data and dependency design

The strongest validation for Multi-region data and dependency design combines an allowed test, a denied test and a failure test. Teams should prove normal workload behavior, demonstrate that prohibited behavior is blocked and exercise recovery under a bounded fault. Metrics, logs, audit events and cost records become part of the acceptance package.

Service and Capability Map

Capability	Service or pattern	Enterprise use
Zonal resilience	Availability Zones	Distribute supported resources across datacenter zones.
Global HTTP	Azure Front Door	Provide global routing, acceleration and web protection.

Capability	Service or pattern	Enterprise use
DNS traffic	Traffic Manager	Route endpoints using DNS-based policies.
VM recovery	Azure Site Recovery	Replicate and orchestrate supported server failover.
Backup	Recovery Services vault / Backup vault	Retain and restore protected data.
Database continuity	Geo-replication / failover groups	Recover supported data services across regions.

Architecture and Operating Decisions

- Use Azure managed services only after recording availability, security, portability and cost consequences.
- Separate production, nonproduction, security and shared services with strong administrative and billing boundaries.
- Prefer short-lived workload identity, private service access and centralized audit evidence over persistent credentials and public administration.
- Automate azure site recovery and backup-based recovery with reviewed plans, policy tests, staged rollout and post-deployment verification.
- Define recovery objectives, quotas, ownership and decommissioning before the service becomes business critical.

Reference Configuration or Decision Pattern

```
recovery_tiers:
  mission_critical: active_active_or_active_passive_multi_region
  business_critical: warm_standby_with_geo_replication
  standard: backup_restore
runbook:
  - declare_incident
  - validate_secondary_dependencies
  - fail_data_then_application_then_traffic
  - verify_integrity
  - controlled_failback
```

Interpretation: Treat the pattern as a starting point. Replace placeholders with measured requirements, provider limits and approved organizational standards.

Production Readiness Checklist

- ☐ Architecture diagrams and responsibility boundaries for azure reliability, global traffic and disaster recovery are current and version controlled.
- ☐ Identity uses groups, roles, workload federation and privileged elevation with no unnecessary standing administrator access.
- ☐ Network paths, DNS behavior, encryption, audit logging and service limits are tested before production use.
- ☐ Changes are delivered through IaC with validation, policy, approval, rollback and drift detection.
- ☐ Recovery, cost allocation, ownership metadata and teardown have measurable evidence.

Troubleshooting Matrix

Symptom	Likely cause	Corrective action	Validation evidence
Deployment succeeds but service is unreachable	Route, firewall, DNS or private-endpoint path is incomplete	Trace the path hop by hop; compare route, policy, DNS and endpoint state; then run an explicit reachability test.	Successful synthetic request and flow/audit evidence
Authorization is broader than intended	Inherited role, wildcard permission or unmanaged credential	Inspect effective permissions, credential age and access logs; replace broad grants with scoped roles and short-lived identity.	Effective-access review and denied-action test
Regional failure breaks the workload	State, dependency or failover mechanism remained single-region	Map stateful dependencies, restore order and traffic controls; exercise failover and fallback against measured RTO/RPO.	Timed recovery log with data-integrity checks

Symptom	Likely cause	Corrective action	Validation evidence
Cost rises without traffic growth	Idle capacity, data transfer, logging volume or commitment mismatch	Break down spend by owner, service, region and usage driver; validate rightsizing or commitment changes against SLOs.	Cost delta plus performance and reliability validation

Hands-on Lab 27: Run an Azure Regional Failover and Failback Exercise

Prerequisites

- Disposable provider sandbox or a local simulation.
- Terraform or provider CLI, Git and a diagramming tool.
- Synthetic identities and data; no production credentials or confidential information.

Implementation

1. Create the architecture and responsibility model for azure reliability, global traffic and disaster recovery using fictional enterprise requirements.
2. Implement the smallest safe sandbox using Availability Zones, Azure Front Door, Traffic Manager; use free or low-cost options where available.
3. Apply identity, network, encryption, logging, metadata and budget controls through IaC or an equivalent version-controlled simulation.
4. Run one approved behavior test and one denied behavior test; save the commands, timestamps and resulting telemetry.
5. Inject a bounded configuration or dependency failure, follow the runbook, recover and record measured recovery time.
6. Destroy or quarantine the lab, revoke temporary access and verify that no unexpected billable resources remain.

Validation Evidence

- Version-controlled architecture, IaC and decision record.
- Allowed, denied and failure-test outputs with timestamps.
- Cost, recovery, ownership and teardown verification.

Cleanup

- Destroy test resources in dependency order and verify the provider inventory and billing view.
- Revoke temporary identities, remove local state copies and sanitize all portfolio evidence.

Interview Questions and Model Answers

1. What is the most important design principle in Azure Reliability, Global Traffic and Disaster Recovery?

Make responsibility and failure boundaries explicit, then automate controls and prove them with normal, denied and recovery tests rather than trusting a successful deployment alone.

2. How would you choose between the main Azure services in this domain?

Start with workload requirements for latency, state, scale, compliance, team capability, recovery and cost. Select the managed service whose operating model fits those requirements and document the consequences in an ADR.

3. What evidence would you require before production approval?

Architecture and threat models, reviewed IaC plan, effective-access results, network and encryption tests, known audit events, SLO and recovery validation, ownership metadata, budget routing and rollback evidence.

4. How do you reduce provider lock-in without creating a lowest-common-denominator platform?

Preserve portable source, data, interfaces and deployment artifacts where a credible exit scenario requires them, while still using provider-specific capabilities when their measurable value exceeds migration and dependency risk.

Chapter Summary

- Availability zones and zone-redundant services defines a key architecture boundary.
- Azure Front Door, Traffic Manager and global routing must be automated and least-privileged.
- Azure Site Recovery and backup-based recovery requires policy, telemetry and recovery evidence.
- Multi-region data and dependency design should be tested before production and after material change.

Chapter 28: Azure Monitor, Application Insights and Hybrid Operations

Building operational visibility across Azure and connected infrastructure

Why this chapter matters: Logs, metrics, traces, resource health and change data must be correlated into actionable operational views. Collection design also needs retention and cost governance.

Learning Objectives

- Explain the architecture, responsibility boundaries and major trade-offs of azure monitor metrics and log analytics.
- Design application insights and opentelemetry for enterprise scale, least privilege and bounded failure.
- Implement azure arc and hybrid inventory through version-controlled automation and policy checks.
- Troubleshoot automation, update management and incident evidence using provider-native telemetry and repeatable evidence.
- Deliver a sanitized portfolio artifact for create an azure operations workbook and incident investigation.

Core Concepts

Azure Monitor metrics and Log Analytics

Azure Monitor metrics and Log Analytics is a design boundary, not merely a console feature. In Azure Monitor, Application Insights and Hybrid Operations, the team must document ownership, trust assumptions, dependency direction, failure impact and the evidence used to prove the control. The recommended implementation uses Azure Monitor Metrics, Log Analytics workspaces, Application Insights, Azure Arc-enabled servers / Kubernetes only where they satisfy an explicit requirement; equivalent services are not automatically interchangeable.

Application Insights and OpenTelemetry

A production implementation of Application Insights and OpenTelemetry separates the management plane from workload execution and constrains blast radius. Architecture reviews should evaluate quotas, regional behavior, identity propagation, private connectivity, encryption, logging and recovery. The operational contract must state who changes the control, who monitors it and how an emergency exception expires.

Azure Arc and hybrid inventory

For Azure Arc and hybrid inventory, repeatability matters more than a one-time successful deployment. Infrastructure as code, policy tests, deployment verification, drift detection and rollback evidence should be retained together. A design is incomplete when it works only for the original engineer or depends on undocumented console steps.

Automation, update management and incident evidence

The strongest validation for Automation, update management and incident evidence combines an allowed test, a denied test and a failure test. Teams should prove normal workload behavior, demonstrate that prohibited behavior is blocked and exercise recovery under a bounded fault. Metrics, logs, audit events and cost records become part of the acceptance package.

Service and Capability Map

Capability	Service or pattern	Enterprise use
Metrics	Azure Monitor Metrics	Collect platform and custom metrics.
Logs	Log Analytics workspaces	Query resource and application logs.
APM	Application Insights	Monitor application dependencies, traces and failures.

Capability	Service or pattern	Enterprise use
Hybrid	Azure Arc-enabled servers / Kubernetes	Project external resources into Azure management.
Change	Azure Update Manager / Automation	Assess and orchestrate supported operational changes.
Health	Azure Resource Health / Service Health	Detect resource and provider incidents.

Architecture and Operating Decisions

- Use Azure managed services only after recording availability, security, portability and cost consequences.
- Separate production, nonproduction, security and shared services with strong administrative and billing boundaries.
- Prefer short-lived workload identity, private service access and centralized audit evidence over persistent credentials and public administration.
- Automate azure arc and hybrid inventory with reviewed plans, policy tests, staged rollout and post-deployment verification.
- Define recovery objectives, quotas, ownership and decommissioning before the service becomes business critical.

Reference Configuration or Decision Pattern

```
observability:
  workspace_strategy: region_data_residency_and_access
  telemetry: [metrics, logs, traces, activity_logs]
  application: opentelemetry_to_application_insights
  alerts: symptom_based_action_groups
  hybrid_inventory: azure_arc
  retention: per_data_class_and_cost_model
```

Interpretation: Treat the pattern as a starting point. Replace placeholders with measured requirements, provider limits and approved organizational standards.

Production Readiness Checklist

- ☐ Architecture diagrams and responsibility boundaries for azure monitor, application insights and hybrid operations are current and version controlled.
- ☐ Identity uses groups, roles, workload federation and privileged elevation with no unnecessary standing administrator access.
- ☐ Network paths, DNS behavior, encryption, audit logging and service limits are tested before production use.
- ☐ Changes are delivered through IaC with validation, policy, approval, rollback and drift detection.
- ☐ Recovery, cost allocation, ownership metadata and teardown have measurable evidence.

Troubleshooting Matrix

Symptom	Likely cause	Corrective action	Validation evidence
Deployment succeeds but service is unreachable	Route, firewall, DNS or private-endpoint path is incomplete	Trace the path hop by hop; compare route, policy, DNS and endpoint state; then run an explicit reachability test.	Successful synthetic request and flow/audit evidence
Authorization is broader than intended	Inherited role, wildcard permission or unmanaged credential	Inspect effective permissions, credential age and access logs; replace broad grants with scoped roles and short-lived identity.	Effective-access review and denied-action test
Regional failure breaks the workload	State, dependency or failover mechanism remained single-region	Map stateful dependencies, restore order and traffic controls; exercise failover and fallback against measured RTO/RPO.	Timed recovery log with data-integrity checks

Symptom	Likely cause	Corrective action	Validation evidence
Cost rises without traffic growth	Idle capacity, data transfer, logging volume or commitment mismatch	Break down spend by owner, service, region and usage driver; validate rightsizing or commitment changes against SLOs.	Cost delta plus performance and reliability validation

Hands-on Lab 28: Create an Azure Operations Workbook and Incident Investigation

Prerequisites

- Disposable provider sandbox or a local simulation.
- Terraform or provider CLI, Git and a diagramming tool.
- Synthetic identities and data; no production credentials or confidential information.

Implementation

1. Create the architecture and responsibility model for azure monitor, application insights and hybrid operations using fictional enterprise requirements.
2. Implement the smallest safe sandbox using Azure Monitor Metrics, Log Analytics workspaces, Application Insights; use free or low-cost options where available.
3. Apply identity, network, encryption, logging, metadata and budget controls through IaC or an equivalent version-controlled simulation.
4. Run one approved behavior test and one denied behavior test; save the commands, timestamps and resulting telemetry.
5. Inject a bounded configuration or dependency failure, follow the runbook, recover and record measured recovery time.
6. Destroy or quarantine the lab, revoke temporary access and verify that no unexpected billable resources remain.

Validation Evidence

- Version-controlled architecture, IaC and decision record.
- Allowed, denied and failure-test outputs with timestamps.
- Cost, recovery, ownership and teardown verification.

Cleanup

- Destroy test resources in dependency order and verify the provider inventory and billing view.
- Revoke temporary identities, remove local state copies and sanitize all portfolio evidence.

Interview Questions and Model Answers

1. What is the most important design principle in Azure Monitor, Application Insights and Hybrid Operations?

Make responsibility and failure boundaries explicit, then automate controls and prove them with normal, denied and recovery tests rather than trusting a successful deployment alone.

2. How would you choose between the main Azure services in this domain?

Start with workload requirements for latency, state, scale, compliance, team capability, recovery and cost. Select the managed service whose operating model fits those requirements and document the consequences in an ADR.

3. What evidence would you require before production approval?

Architecture and threat models, reviewed IaC plan, effective-access results, network and encryption tests, known audit events, SLO and recovery validation, ownership metadata, budget routing and rollback evidence.

4. How do you reduce provider lock-in without creating a lowest-common-denominator platform?

Preserve portable source, data, interfaces and deployment artifacts where a credible exit scenario requires them, while still using provider-specific capabilities when their measurable value exceeds migration and dependency risk.

Chapter Summary

- Azure Monitor metrics and Log Analytics defines a key architecture boundary.
- Application Insights and OpenTelemetry must be automated and least-privileged.
- Azure Arc and hybrid inventory requires policy, telemetry and recovery evidence.
- Automation, update management and incident evidence should be tested before production and after material change.

Chapter 29: Azure Cost Management, Governance and FinOps

Governing subscriptions, commitments and unit cost

Why this chapter matters: Azure cost management must connect subscription ownership and resource metadata to budgets, anomalies, reservations, savings plans and architecture decisions.

Learning Objectives

- Explain the architecture, responsibility boundaries and major trade-offs of cost analysis, exports and allocation.
- Design budgets, anomaly review and ownership for enterprise scale, least privilege and bounded failure.
- Implement reservations and azure savings plan through version-controlled automation and policy checks.
- Troubleshoot advisor recommendations and unit economics using provider-native telemetry and repeatable evidence.
- Deliver a sanitized portfolio artifact for build an azure cost allocation and commitment review.

Core Concepts

Cost analysis, exports and allocation

Cost analysis, exports and allocation is a design boundary, not merely a console feature. In Azure Cost Management, Governance and FinOps, the team must document ownership, trust assumptions, dependency direction, failure impact and the evidence used to prove the control. The recommended implementation uses Cost Management + Billing, Budgets / cost alerts, Azure Advisor, Reservations / Azure savings plan for compute only where they satisfy an explicit requirement; equivalent services are not automatically interchangeable.

Budgets, anomaly review and ownership

A production implementation of Budgets, anomaly review and ownership separates the management plane from workload execution and constrains blast radius. Architecture reviews should evaluate quotas, regional behavior, identity propagation, private connectivity, encryption, logging and recovery. The operational contract must state who changes the control, who monitors it and how an emergency exception expires.

Reservations and Azure savings plan

For Reservations and Azure savings plan, repeatability matters more than a one-time successful deployment. Infrastructure as code, policy tests, deployment verification, drift detection and rollback evidence should be retained together. A design is incomplete when it works only for the original engineer or depends on undocumented console steps.

Advisor recommendations and unit economics

The strongest validation for Advisor recommendations and unit economics combines an allowed test, a denied test and a failure test. Teams should prove normal workload behavior, demonstrate that prohibited behavior is blocked and exercise recovery under a bounded fault. Metrics, logs, audit events and cost records become part of the acceptance package.

Service and Capability Map

Capability	Service or pattern	Enterprise use
Analysis	Cost Management + Billing	Analyze, export and allocate Azure spend.
Control	Budgets / cost alerts	Notify accountable owners before thresholds are exceeded.
Optimization	Azure Advisor	Identify rightsizing, reliability and cost opportunities.
Commitments	Reservations / Azure savings plan for compute	Discount stable eligible usage.

Capability	Service or pattern	Enterprise use
Metadata	Tags / subscription hierarchy	Allocate cost and automate governance.
Chargeback	Cost allocation rules	Distribute shared costs to consuming units.

Architecture and Operating Decisions

- Use Azure managed services only after recording availability, security, portability and cost consequences.
- Separate production, nonproduction, security and shared services with strong administrative and billing boundaries.
- Prefer short-lived workload identity, private service access and centralized audit evidence over persistent credentials and public administration.
- Automate reservations and azure savings plan with reviewed plans, policy tests, staged rollout and post-deployment verification.
- Define recovery objectives, quotas, ownership and decommissioning before the service becomes business critical.

Reference Configuration or Decision Pattern

```
finops:
  allocation: subscription_plus_required_tags
  exports: daily_to_governed_storage
  budgets: owner_and_product_routing
  anomaly_review: weekly
  commitments: baseline_demand_only
  unit_metrics: [cost_per_transaction, cost_per_active_user]
  guardrail: optimization_must_preserve_slo_and_recovery
```

Interpretation: Treat the pattern as a starting point. Replace placeholders with measured requirements, provider limits and approved organizational standards.

Production Readiness Checklist

- ☐ Architecture diagrams and responsibility boundaries for azure cost management, governance and finops are current and version controlled.
- ☐ Identity uses groups, roles, workload federation and privileged elevation with no unnecessary standing administrator access.
- ☐ Network paths, DNS behavior, encryption, audit logging and service limits are tested before production use.
- ☐ Changes are delivered through IaC with validation, policy, approval, rollback and drift detection.
- ☐ Recovery, cost allocation, ownership metadata and teardown have measurable evidence.

Troubleshooting Matrix

Symptom	Likely cause	Corrective action	Validation evidence
Deployment succeeds but service is unreachable	Route, firewall, DNS or private-endpoint path is incomplete	Trace the path hop by hop; compare route, policy, DNS and endpoint state; then run an explicit reachability test.	Successful synthetic request and flow/audit evidence
Authorization is broader than intended	Inherited role, wildcard permission or unmanaged credential	Inspect effective permissions, credential age and access logs; replace broad grants with scoped roles and short-lived identity.	Effective-access review and denied-action test
Regional failure breaks the workload	State, dependency or failover mechanism remained single-region	Map stateful dependencies, restore order and traffic controls; exercise failover and fallback against measured RTO/RPO.	Timed recovery log with data-integrity checks
Cost rises without traffic growth	Idle capacity, data transfer, logging volume or commitment mismatch	Break down spend by owner, service, region and usage driver; validate rightsizing or commitment changes against SLOs.	Cost delta plus performance and reliability validation

Hands-on Lab 29: Build an Azure Cost Allocation and Commitment Review

Prerequisites

- Disposable provider sandbox or a local simulation.
- Terraform or provider CLI, Git and a diagramming tool.
- Synthetic identities and data; no production credentials or confidential information.

Implementation

1. Create the architecture and responsibility model for azure cost management, governance and finops using fictional enterprise requirements.
2. Implement the smallest safe sandbox using Cost Management + Billing, Budgets / cost alerts, Azure Advisor; use free or low-cost options where available.
3. Apply identity, network, encryption, logging, metadata and budget controls through IaC or an equivalent version-controlled simulation.
4. Run one approved behavior test and one denied behavior test; save the commands, timestamps and resulting telemetry.
5. Inject a bounded configuration or dependency failure, follow the runbook, recover and record measured recovery time.
6. Destroy or quarantine the lab, revoke temporary access and verify that no unexpected billable resources remain.

Validation Evidence

- Version-controlled architecture, IaC and decision record.
- Allowed, denied and failure-test outputs with timestamps.
- Cost, recovery, ownership and teardown verification.

Cleanup

- Destroy test resources in dependency order and verify the provider inventory and billing view.
- Revoke temporary identities, remove local state copies and sanitize all portfolio evidence.

Interview Questions and Model Answers

1. What is the most important design principle in Azure Cost Management, Governance and FinOps?

Make responsibility and failure boundaries explicit, then automate controls and prove them with normal, denied and recovery tests rather than trusting a successful deployment alone.

2. How would you choose between the main Azure services in this domain?

Start with workload requirements for latency, state, scale, compliance, team capability, recovery and cost. Select the managed service whose operating model fits those requirements and document the consequences in an ADR.

3. What evidence would you require before production approval?

Architecture and threat models, reviewed IaC plan, effective-access results, network and encryption tests, known audit events, SLO and recovery validation, ownership metadata, budget routing and rollback evidence.

4. How do you reduce provider lock-in without creating a lowest-common-denominator platform?

Preserve portable source, data, interfaces and deployment artifacts where a credible exit scenario requires them, while still using provider-specific capabilities when their measurable value exceeds migration and dependency risk.

Chapter Summary

- Cost analysis, exports and allocation defines a key architecture boundary.
- Budgets, anomaly review and ownership must be automated and least-privileged.
- Reservations and Azure savings plan requires policy, telemetry and recovery evidence.
- Advisor recommendations and unit economics should be tested before production and after material change.

Chapter 30: Azure Enterprise Landing-Zone Capstone

Delivering a governed subscription from request through recovery and closure

Why this chapter matters: An Azure landing zone is successful when the platform team can safely deliver application subscriptions with policy, identity, network, monitoring, security, cost and lifecycle controls.

Learning Objectives

- Explain the architecture, responsibility boundaries and major trade-offs of platform landing-zone services.
- Design application subscription vending for enterprise scale, least privilege and bounded failure.
- Implement workload deployment and reliability tests through version-controlled automation and policy checks.
- Troubleshoot governance, operations and decommissioning evidence using provider-native telemetry and repeatable evidence.
- Deliver a sanitized portfolio artifact for deliver the azure enterprise foundation capstone.

Core Concepts

Platform landing-zone services

Platform landing-zone services is a design boundary, not merely a console feature. In Azure Enterprise Landing-Zone Capstone, the team must document ownership, trust assumptions, dependency direction, failure impact and the evidence used to prove the control. The recommended implementation uses Management groups / subscriptions, Entra ID / PIM / managed identities, Virtual WAN or hub-spoke / Private Link, Policy / Defender / Sentinel only where they satisfy an explicit requirement; equivalent services are not automatically interchangeable.

Application subscription vending

A production implementation of Application subscription vending separates the management plane from workload execution and constrains blast radius. Architecture reviews should evaluate quotas, regional behavior, identity propagation, private connectivity, encryption, logging and recovery. The operational contract must state who changes the control, who monitors it and how an emergency exception expires.

Workload deployment and reliability tests

For Workload deployment and reliability tests, repeatability matters more than a one-time successful deployment. Infrastructure as code, policy tests, deployment verification, drift detection and rollback evidence should be retained together. A design is incomplete when it works only for the original engineer or depends on undocumented console steps.

Governance, operations and decommissioning evidence

The strongest validation for Governance, operations and decommissioning evidence combines an allowed test, a denied test and a failure test. Teams should prove normal workload behavior, demonstrate that prohibited behavior is blocked and exercise recovery under a bounded fault. Metrics, logs, audit events and cost records become part of the acceptance package.

Service and Capability Map

Capability	Service or pattern	Enterprise use
Hierarchy	Management groups / subscriptions	Establish durable governance and workload boundaries.
Identity	Entra ID / PIM / managed identities	Control humans and workloads.
Network	Virtual WAN or hub-spoke / Private Link	Provide governed connectivity.
Security	Policy / Defender / Sentinel	Enforce baseline and central response.
Operations	Azure Monitor / Arc	Monitor Azure and hybrid resources.

Capability	Service or pattern	Enterprise use
Cost	Cost Management / budgets	Allocate and control spend.

Architecture and Operating Decisions

- Use Azure managed services only after recording availability, security, portability and cost consequences.
- Separate production, nonproduction, security and shared services with strong administrative and billing boundaries.
- Prefer short-lived workload identity, private service access and centralized audit evidence over persistent credentials and public administration.
- Automate workload deployment and reliability tests with reviewed plans, policy tests, staged rollout and post-deployment verification.
- Define recovery objectives, quotas, ownership and decommissioning before the service becomes business critical.

Reference Configuration or Decision Pattern

```
acceptance:
  subscription_vending_minutes: 30
  permanent_global_admins: 0
  public_paaS_access_denied: true
  required_diagnostics_coverage_percent: 100
  regional_recovery_tested: true
  cost_owner_coverage_percent: 100
  decommission_residual_resources: 0
```

Interpretation: Treat the pattern as a starting point. Replace placeholders with measured requirements, provider limits and approved organizational standards.

Production Readiness Checklist

- ☐ Architecture diagrams and responsibility boundaries for azure enterprise landing-zone capstone are current and version controlled.
- ☐ Identity uses groups, roles, workload federation and privileged elevation with no unnecessary standing administrator access.
- ☐ Network paths, DNS behavior, encryption, audit logging and service limits are tested before production use.
- ☐ Changes are delivered through IaC with validation, policy, approval, rollback and drift detection.
- ☐ Recovery, cost allocation, ownership metadata and teardown have measurable evidence.

Troubleshooting Matrix

Symptom	Likely cause	Corrective action	Validation evidence
Deployment succeeds but service is unreachable	Route, firewall, DNS or private-endpoint path is incomplete	Trace the path hop by hop; compare route, policy, DNS and endpoint state; then run an explicit reachability test.	Successful synthetic request and flow/audit evidence
Authorization is broader than intended	Inherited role, wildcard permission or unmanaged credential	Inspect effective permissions, credential age and access logs; replace broad grants with scoped roles and short-lived identity.	Effective-access review and denied-action test
Regional failure breaks the workload	State, dependency or failover mechanism remained single-region	Map stateful dependencies, restore order and traffic controls; exercise failover and fallback against measured RTO/RPO.	Timed recovery log with data-integrity checks
Cost rises without traffic growth	Idle capacity, data transfer, logging volume or commitment mismatch	Break down spend by owner, service, region and usage driver; validate rightsizing or commitment changes against SLOs.	Cost delta plus performance and reliability validation

Hands-on Lab 30: Deliver the Azure Enterprise Foundation Capstone

Prerequisites

- Disposable provider sandbox or a local simulation.
- Terraform or provider CLI, Git and a diagramming tool.
- Synthetic identities and data; no production credentials or confidential information.

Implementation

1. Create the architecture and responsibility model for azure enterprise landing-zone capstone using fictional enterprise requirements.
2. Implement the smallest safe sandbox using Management groups / subscriptions, Entra ID / PIM / managed identities, Virtual WAN or hub-spoke / Private Link; use free or low-cost options where available.
3. Apply identity, network, encryption, logging, metadata and budget controls through IaC or an equivalent version-controlled simulation.
4. Run one approved behavior test and one denied behavior test; save the commands, timestamps and resulting telemetry.
5. Inject a bounded configuration or dependency failure, follow the runbook, recover and record measured recovery time.
6. Destroy or quarantine the lab, revoke temporary access and verify that no unexpected billable resources remain.

Validation Evidence

- Version-controlled architecture, IaC and decision record.
- Allowed, denied and failure-test outputs with timestamps.
- Cost, recovery, ownership and teardown verification.

Cleanup

- Destroy test resources in dependency order and verify the provider inventory and billing view.
- Revoke temporary identities, remove local state copies and sanitize all portfolio evidence.

Interview Questions and Model Answers

1. What is the most important design principle in Azure Enterprise Landing-Zone Capstone?

Make responsibility and failure boundaries explicit, then automate controls and prove them with normal, denied and recovery tests rather than trusting a successful deployment alone.

2. How would you choose between the main Azure services in this domain?

Start with workload requirements for latency, state, scale, compliance, team capability, recovery and cost. Select the managed service whose operating model fits those requirements and document the consequences in an ADR.

3. What evidence would you require before production approval?

Architecture and threat models, reviewed IaC plan, effective-access results, network and encryption tests, known audit events, SLO and recovery validation, ownership metadata, budget routing and rollback evidence.

4. How do you reduce provider lock-in without creating a lowest-common-denominator platform?

Preserve portable source, data, interfaces and deployment artifacts where a credible exit scenario requires them, while still using provider-specific capabilities when their measurable value exceeds migration and dependency risk.

Chapter Summary

- Platform landing-zone services defines a key architecture boundary.
- Application subscription vending must be automated and least-privileged.
- Workload deployment and reliability tests requires policy, telemetry and recovery evidence.
- Governance, operations and decommissioning evidence should be tested before production and after material change.

Azure Enterprise Architecture and Operations Reference

Use this reference to review architecture outcomes, implementation controls and the evidence required for production approval.

Domain	Required outcome	Validation evidence
Hierarchy and ownership	Durable resource boundaries with accountable owners	Inventory, effective policy and lifecycle tests
Identity	Federated humans and keyless workloads	Token, role, access-review and denied-action evidence
Network	Approved routes, DNS, private access and egress	Reachability, flow, DNS, latency and failover tests
Data	Classified, encrypted and recoverable state	Backup, restore, reconciliation and exit evidence
Security	Preventive, detective and recovery controls	Policy results, findings, containment and audit timeline
Operations	SLOs, telemetry, runbooks and incident process	Dashboards, alerts, drill timeline and postmortem
FinOps	Allocation, budgets, unit economics and commitment governance	Billing export, owner coverage and optimization record
Delivery	Reviewed IaC and immutable artifact promotion	Plan, policy, signature, deployment and rollback chain
Reliability	Measured capacity, RTO/RPO and failback	Load, failure, recovery and integrity tests
Lifecycle	Vending, upgrade, exception and closure	Fresh deployment, drift and zero-residual teardown

Completion Checklist

- ☐ I can explain the architecture and trade-offs in every chapter of 12C.
- ☐ I can implement at least one sandbox pattern through IaC and short-lived identity.
- ☐ I can demonstrate approved, denied and failure behavior with provider-native evidence.
- ☐ I can design private connectivity, data protection, recovery and cost governance.
- ☐ I can produce architecture diagrams, ADRs, runbooks and troubleshooting timelines.
- ☐ I can destroy the lab safely and prove that no unexpected billable resource remains.

Additional Portfolio Projects

1. Resource vending workflow with ownership, policy, budget and closure automation.
2. Keyless CI/CD identity module and effective-access test harness.
3. Private network and DNS validation pack with route and failure scenarios.
4. Managed data backup, restore and reconciliation project.
5. Security finding triage and automated containment simulation.
6. SLO dashboard and incident-command game day.
7. Billing export, allocation coverage and unit-cost dashboard.
8. IaC provider-upgrade and drift-remediation pipeline.
9. Regional or cross-provider disaster-recovery exercise.
10. Executive architecture brief with decisions, risks, roadmap and measured outcomes.

30-Day Practice Plan

Days	Focus	Required output
1-4	Hierarchy, ownership and identity	Architecture diagram, RACI and access tests
5-8	Networking and private services	IP plan, flow matrix and reachability evidence
9-12	Compute and delivery	Workload deployment, autoscaling and rollback
13-16	Data and recovery	Backup, restore and integrity report
17-20	Security and audit	Threat model, policy tests and finding workflow
21-23	Observability and incidents	SLO, dashboard and incident timeline
24-26	FinOps and governance	Allocation, budget and unit-cost review
27-28	Failure drills	Bounded outage and recovery evidence
29-30	Capstone and portfolio	Architecture defense, cleanup and sanitized report

Glossary

Term	Definition
Application landing zone	A governed environment in which a workload team deploys and operates applications.
Blast radius	The maximum scope of impact from a failure, credential or change.
Control plane	Interfaces and services used to administer a cloud or platform.
Data plane	The runtime path that processes application traffic and data.
Federation	Trust that allows an external identity source to authenticate without duplicate long-lived credentials.
Guardrail	A preventive, detective or corrective control that constrains activity.
Landing zone	A scalable cloud foundation for hierarchy, identity, network, security, logging and cost.
Private endpoint	A private network interface used to access a managed service without a public path.
RPO	Maximum acceptable data loss measured in time.
RTO	Target time to restore a service after disruption.
Unit cost	Technology cost per useful business or technical output.
Workload identity	A short-lived identity issued to software based on trusted runtime context.

Official Documentation Index

The following official sources were checked for this 2026 edition. Verify current service behavior, quotas, prices and certification requirements before production use or exam registration.

- [Azure Landing Zones](#)
- [Azure Well-Architected Framework](#)
- [Microsoft Entra Architecture](#)
- [Azure Hub-Spoke Architecture](#)
- [Azure Private Link](#)
- [Microsoft Defender for Cloud](#)
- [Azure Monitor Overview](#)
- [Azure Cost Management](#)

Volume 12C completion: preserve your architecture, code, test evidence, recovery timeline, cost review and sanitized final report as one portfolio package.